

CSE400 - Fundamentals of Probability in computing

Lecture 3

Lecture scribe

Contents

1	Introduction	3
2	Understanding the CSE400 Course Project	3
2.1	Project Weightage	3
2.2	Team Formation	3
2.3	Deadline for Team Formation	3
3	Project Execution Guidelines	4
3.1	Milestones	4
3.2	Artifacts to be Produced	4
3.3	Deliverables	4
3.4	Project Assessment	5
4	Project Milestones (M1 – M6)	5
4.1	Milestone 1 (M1): Kickstart	5
4.2	Milestone 2 (M2): Mathematical Modeling	6
4.3	Milestone 3 (M3): Coding	6
4.4	Milestone 4 (M4): Inference	7
4.5	Milestone 5 (M5): Randomized Algorithms	7
4.6	Milestone 6 (M6): Bounds and Analysis	7
5	Submission 1 – Concept Evolution Maps	8

6	Submission 2 – Scribe (Learning Reflection Logs)	8
6.1	Lecture Scribe	8
6.2	Project Scribe	8
6.3	Trade-Off Analysis	9
7	Submission 3 – Multimodal Artifacts	9
7.1	Video Requirements	9
7.2	Video Content	9
7.3	Types of Multimedia Artifacts	9
8	Summary	10

1 Introduction

The third lecture of the CSE400 course focuses on the initiation and execution of the course project. The project component is an important part of the course and is designed to help students apply theoretical concepts to real-world problems. Through this project, students are expected to engage in problem identification, mathematical modeling, coding, and analytical evaluation.

The lecture primarily introduces the structure of the project, its milestones, the types of submissions required, and the evaluation process. It also explains the artifacts students must create during the project lifecycle, including concept evolution maps, scribes, and multimedia artifacts.

The goal of the project is to ensure that students not only understand theoretical concepts but also develop the ability to apply them in a structured and research-oriented manner.

2 Understanding the CSE400 Course Project

The CSE400 course includes a project component that encourages collaborative work, analytical thinking, and practical implementation of concepts learned during the course.

2.1 Project Weightage

The project contributes **30% of the total course evaluation**. This makes the project a significant part of the course assessment.

2.2 Team Formation

Students are required to form teams to work on the project. Team collaboration plays a crucial role in completing the various milestones and deliverables associated with the project.

2.3 Deadline for Team Formation

The deadline for forming teams is:

17 January 2026 (Saturday – End of Day)

After teams are formed, students begin working on the project according to the milestone structure defined in the course.

3 Project Execution Guidelines

The project is carried out through a series of milestones and structured submissions.

3.1 Milestones

The project consists of **six milestones**, labeled from **M1 to M6**. Each milestone represents a specific phase in the project development process.

Each team is required to submit **one submission per milestone**.

3.2 Artifacts to be Produced

During the project, students must produce several types of artifacts, including:

- Concept Evolution Maps
- Scribe Documentation
- Multimodal Artifacts
- Question-Driven Artifacts
- Collaboration and Team Dynamics Artifacts

These artifacts document the progress and evolution of the project throughout the semester.

3.3 Deliverables

The project deliverables may include:

- Code implementations
- Reports
- Videos
- Other artifacts specified during the course

3.4 Project Assessment

The evaluation of the project includes several components:

- Team assessment before the mid-semester examination
- Team assessment after the mid-semester examination
- Final project viva
- Final submission at the end of the course

These assessments help measure the progress and understanding of the project work.

4 Project Milestones (M1 – M6)

The project development process is divided into six milestones, each focusing on a particular stage of the project.

4.1 Milestone 1 (M1): Kickstart

The first milestone focuses on initiating the project and defining the problem.

Activities involved in this milestone include:

1. Team Formation
2. Area Identification
3. Background Study
4. Motivation
5. Problem Formulation

Students first form teams and then identify a suitable area for their project. After selecting the area, they conduct a background study to understand existing knowledge related to the problem. The motivation behind choosing the problem must also be clearly explained.

Finally, the problem is formulated in a clear and structured way so that it can be analyzed and solved in later stages.

4.2 Milestone 2 (M2): Mathematical Modeling

In this milestone, the real-world problem identified earlier is converted into a mathematical representation.

The mathematical model may involve several probabilistic concepts, including:

- Random Variables (RV)
- Probability Mass Function (PMF)
- Probability Density Function (PDF)
- Cumulative Distribution Function (CDF)
- Multivariate Random Variables
- Joint PMF
- Joint PDF
- Joint CDF

Mathematical modeling helps represent complex real-world problems in a structured mathematical form. This makes it possible to analyze the problem using computational techniques.

4.3 Milestone 3 (M3): Coding

After mathematical modeling, the next stage involves implementing the model using programming.

The tasks in this milestone include:

- Implementing the mathematical model using code
- Performing simulations
- Conducting computational analysis

The purpose of this stage is to verify whether the mathematical model behaves as expected when implemented computationally.

4.4 Milestone 4 (M4): Inference

This milestone focuses on the use of randomized algorithms.

Students are required to:

1. Select a randomized algorithm
2. Understand how the algorithm works
3. Implement the algorithm using code

Randomized algorithms incorporate randomness as part of their logic. These algorithms can sometimes provide efficient solutions to problems that are difficult to solve deterministically.

4.5 Milestone 5 (M5): Randomized Algorithms

After implementing the randomized algorithm, students apply it to their project problem.

The main tasks include:

- Applying the randomized algorithm to the problem domain
- Analyzing the results produced by the algorithm

Students may also compare the performance of randomized algorithms with deterministic algorithms to observe differences and gain insights.

4.6 Milestone 6 (M6): Bounds and Analysis

The final milestone focuses on theoretical analysis and final submission.

Activities in this milestone include:

- Deriving theoretical bounds
- Analyzing the results obtained from the algorithms
- Compiling the findings
- Preparing the final submission

The final submission may include:

- Code
- Reports
- Videos
- Supporting artifacts

5 Submission 1 – Concept Evolution Maps

Concept evolution maps are used to visualize how the project idea develops over time.

Students may use tools such as:

- Miro Concept Map Tool
- Draw.io (diagrams.net)

These maps help represent the relationship between different ideas and show how the project concept evolves throughout the semester.

6 Submission 2 – Scribe (Learning Reflection Logs)

Students are required to prepare scribes documenting lecture learnings and project-related decisions.

There are two types of scribes.

6.1 Lecture Scribe

Lecture scribes summarize the content discussed during lectures.

- Two groups are assigned to prepare a lecture scribe.
- The scribe should clearly explain the lecture content.
- The minimum length required is **8–10 pages**.

6.2 Project Scribe

Project scribes document the decision-making process during project development.

They include:

- Decision logs
- Explanation of why one option was chosen over another
- Constraints faced
- Alternatives considered
- Final decisions taken
- Supporting evidence

6.3 Trade-Off Analysis

Students may use trade-off matrices to compare different alternatives based on factors such as:

- Cost
- Performance
- Risk

These comparisons help justify the decisions made during the project.

7 Submission 3 – Multimodal Artifacts

Students must also produce multimedia artifacts to explain their project work.

7.1 Video Requirements

- One video must be submitted for each milestone.
- The video duration should be **10–15 minutes**.

7.2 Video Content

The video should explain:

- The work completed in the milestone
- Coding and simulation results if applicable

Students may use tools such as PowerPoint, Google Slides, or screen recording software to create these videos.

7.3 Types of Multimedia Artifacts

Examples include:

- Think-Aloud Videos
- One-Minute Insight Videos
- Project Demonstration Videos

The evaluation focuses mainly on the quality of the content rather than video editing skills.

8 Summary

Lecture 3 introduces the overall framework for executing the CSE400 course project. The lecture explains the project structure, milestone-based execution process, required artifacts, and evaluation criteria.

The project contributes 30% to the overall course grade and is divided into six milestones that guide students through problem formulation, mathematical modeling, coding, algorithm implementation, and analysis.

Students are also required to prepare concept evolution maps, lecture and project scribes, and multimedia artifacts to document their learning and project development process.

Overall, the lecture emphasizes a structured approach to project execution, ensuring that students apply theoretical concepts in practical and analytical contexts.